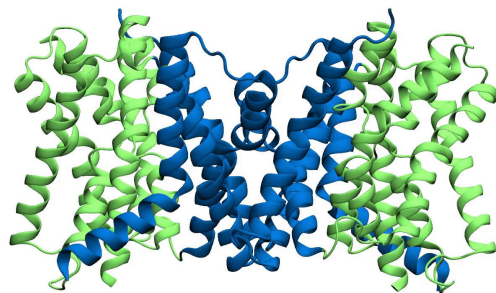


Molecular dynamics simulations



Protein Science course

Lorena Zuzic

21st October 2025

Agenda

Chapter 1: The problem of structural biology

Chapter 2: MD simulations in essence

Chapter 3: Input / Output

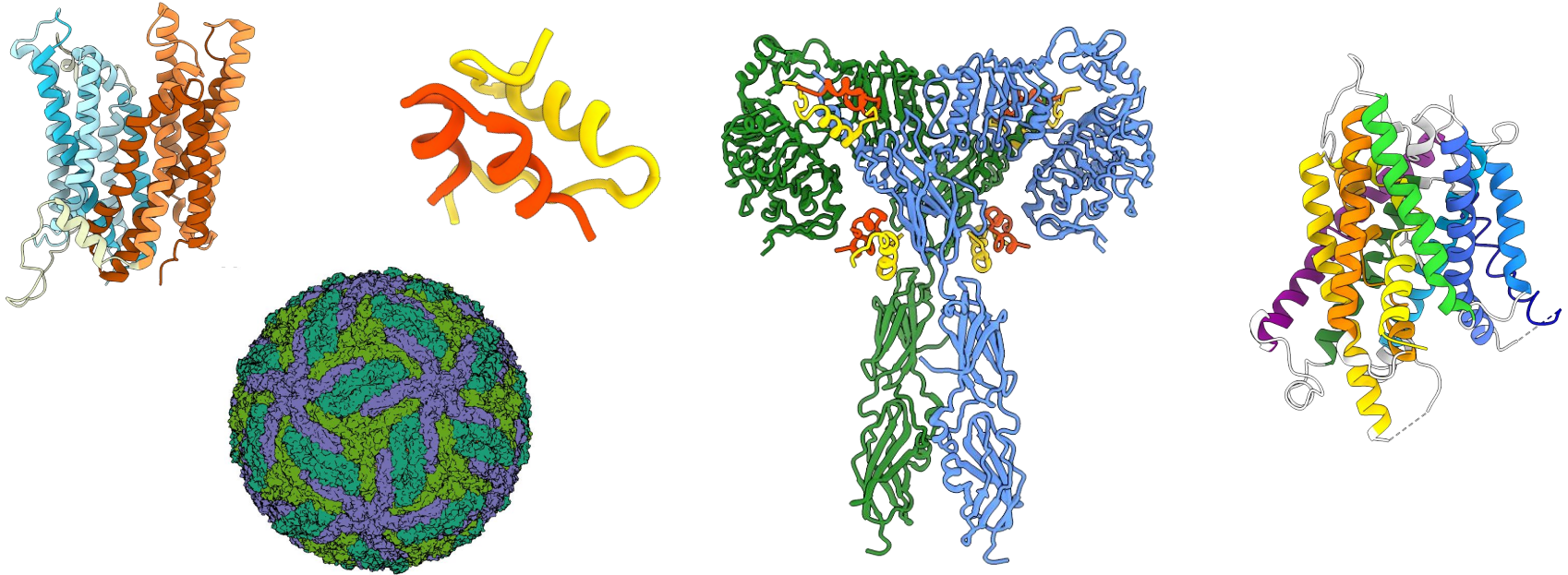
Chapter 4: Applications

Chapter 5: Past, present, future

Chapter 6: Computer exercise

Chapter 1: The problem of structural biology

- Static structures are only snapshots of inherently dynamic entities



Static structures are describing dynamic processes

Protein function is often 'encoded' in its movements

- Proteins as molecular machines

Structural biology generates snapshots

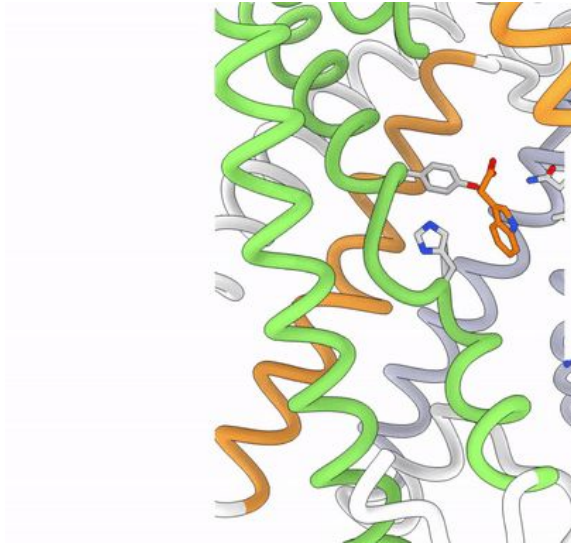
- It is inherently difficult to understand the mechanism of machines using only one (or several) pictures

Other processes where dynamics are relevant:

- Ligand binding/unbinding, lipid organisation in a membrane, phase-separation, substrate transport, proton pumping, protein-protein interactions (complex formation), ...

→ **molecular dynamics**

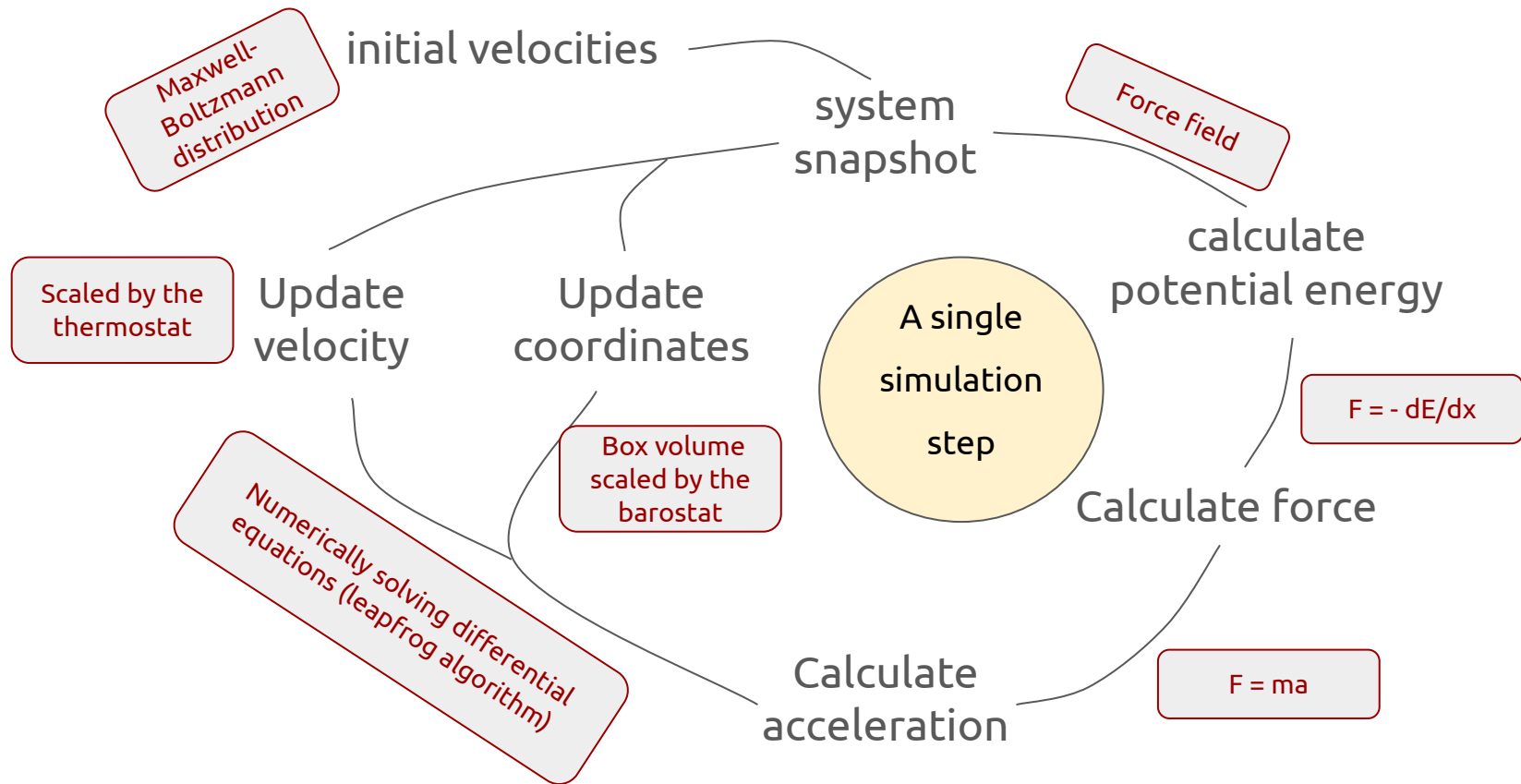
Molecular dynamics (MD) simulations



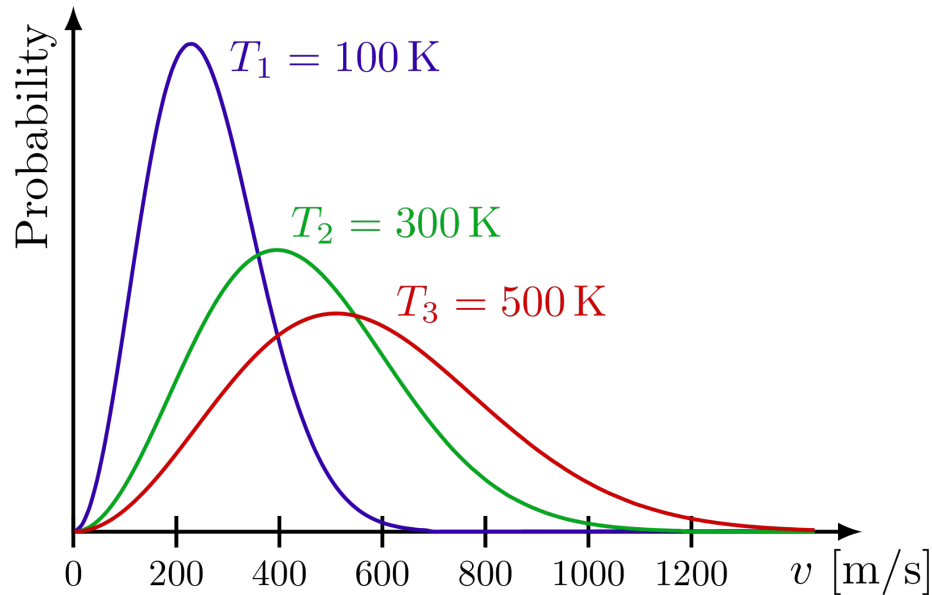
“a computational microscope”

- “**Molecular dynamics (MD)** is a [computer simulation](#) method for analyzing the [physical movements](#) of [atoms](#) and [molecules](#)”. (Wikipedia)
- Based on physical principles!
- The movement is the result of **calculations - numerically solving Newton's equations of motions**
- Atoms are balls, bonds are springs

Simulations proceed in a stepwise manner



Maxwell-Boltzmann distribution



- The initial velocities are randomly distributed to each particle in the system
- Distribution of velocities is taken from a Maxwell-Boltzmann distribution (temperature dependent)
- Initial **velocities reflect system temperature**

Force fields

A **force field** is a collection of interatomic potentials; in practice, a collection of functions and parameters used to calculate the potential energy of the system.

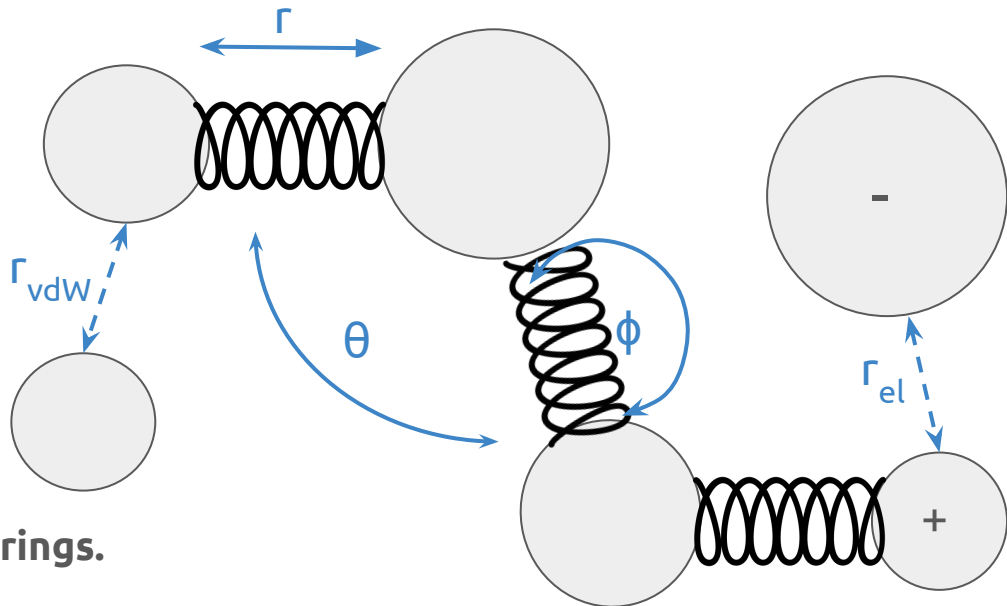
Atoms are balls and covalent bonds are springs.

Electrons are implicit (in bonds and partial charges) - bond creation or bond breaking is **not allowed**.

$$E = E_{\text{bond}} + E_{\text{angle}} + E_{\text{dihedral}} + E_{\text{vdW}} + E_{\text{electrostatic}}$$

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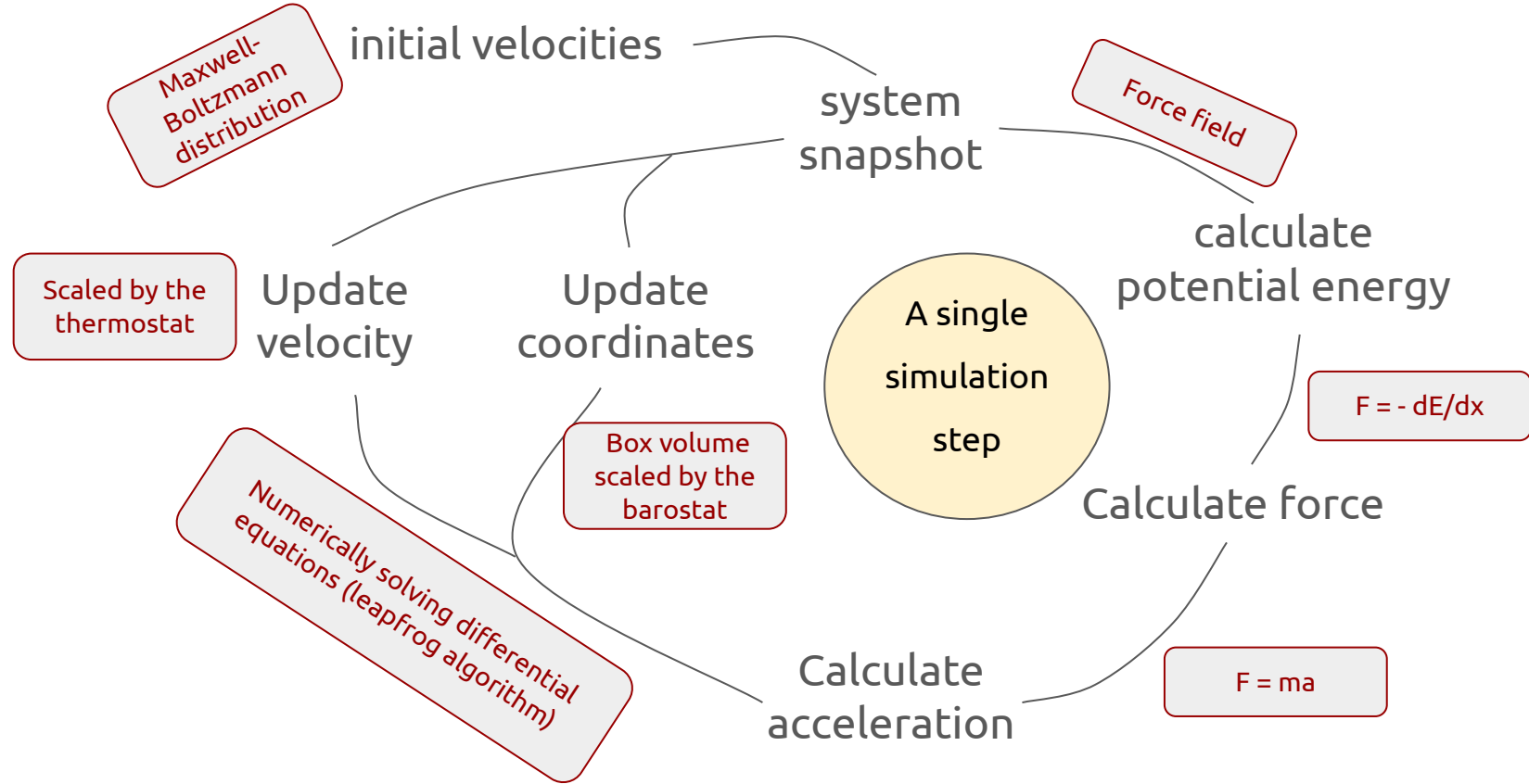
bonded terms non-bonded terms



Molecular mechanics

$$E_{\text{bond}} = K_{\text{bond}} (r - r_0)^2$$

FF flavours: Charmm, Amber, ...



How long is a single simulation step?

- **If too short**, the simulations would take an unfeasibly long time to run
- **If too long**, the simulations would be inaccurate.

The Goldilocks zone: **2 fs**

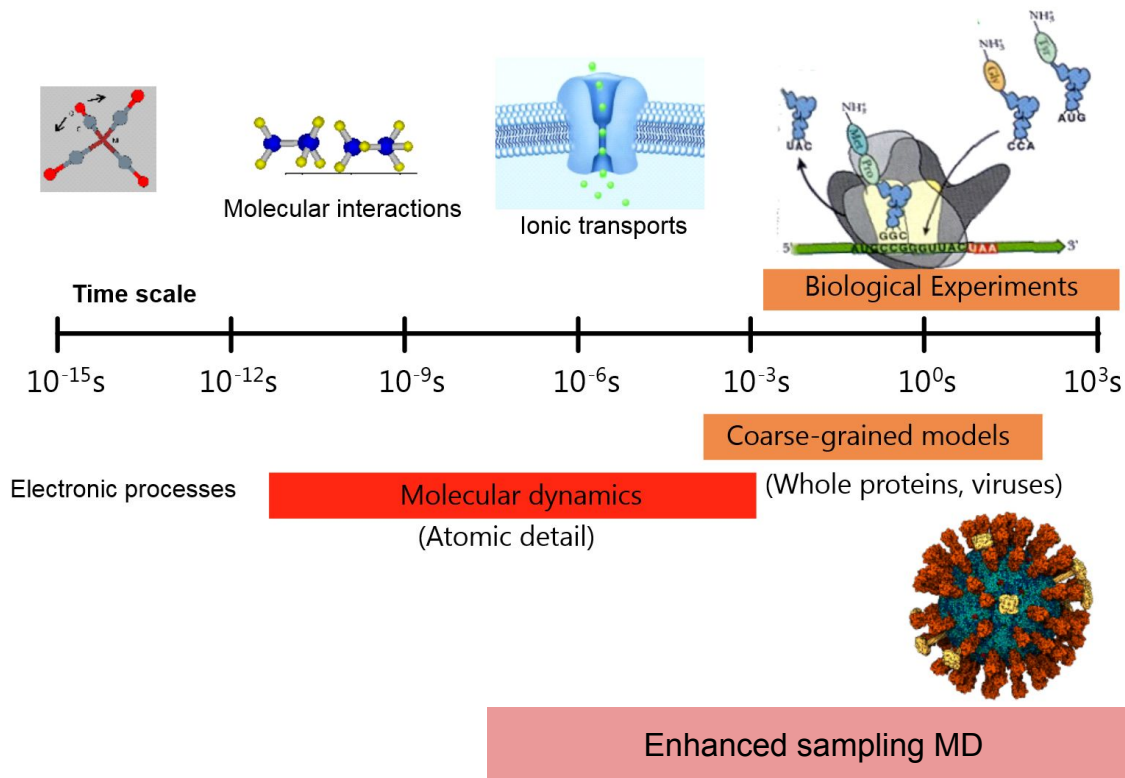
- The step is limited by the highest frequency motion in the system

Number of steps in a typical simulation: 10^8 - 10^{10}

Typical simulation time: 0.1 - 10 μ s

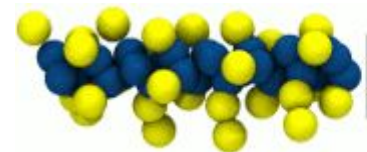
The length of a simulation is dependent on the system size and complexity!

MD simulations can bridge the time gap



Coarse-grained methods:

Atoms are grouped together into "beads"



Enhanced sampling methods:

enhance sampling of rare states
(e.g., conformational transitions)

Chapter 3: Input/output

Input

- **Coordinate file:** containing all initial atoms of a system
- **Topology file(s):** specifying all bonds, angles, dihedrals in the system
- **Force field files**
- **Parameter file:** defining simulation conditions

Engine

- A simulation software that performs all the calculations
 - on high-performance computing (HPC) cluster
 - Simulations running for days / weeks / months

Output

- **Trajectory file:** positions of all atoms over (saved) simulation steps
- **Log file:** logging the simulation metadata
- **Checkpoint file:** allows restarting from that checkpoint
- **Energy file:** information about energies, volume, temperature, pressure...

Input: coordinate file

- A .pdb file or similar
- Contains coordinate information for all elements of the system

Protein

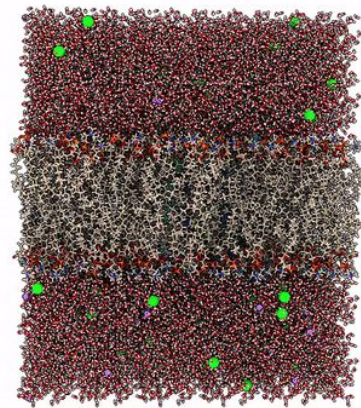
- Needs to be patched (if short loops are missing)
- Adding post-translational modifications
- Ligand treatment
- Setting up protonation states

Environment

- E.g., membrane: composition needs to be chosen
- Solvent: usually water and 0.15 NaCl
- Other solutes or ligands...

Simulation box

- Periodic boundary conditions
- Particle should *not* see the mirror image of itself



Strict control of the environment

- A position and identity of every single atom in the system is controlled by the researcher

Reflection of the native environment

- Native membrane composition
- Control over post-translational modifications
- Mimicking pH with protonation states
- No experimental additives (detergents, excipients, buffer components...)

Experimentally impossible setups

- Setting up conditions that are experimentally impossible
 - Testing protonation of only one residue
- Adding “unphysical” elements
 - E.g., ligands that only interact with protein, but not with solvent
- Alchemy is allowed (if it serves a purpose)

Input: parameter file

```
integrator      = md
dt             = 0.002
nsteps         = 500000000      ; 1000 ns
nstxout-compressed = 50000
nstxtcout      = 5000          ; .xtc output, every 10 ps
nstvout        = 0            ; velocity output
nstxout        = 0            ; .trr output
nstenergy      = 5000         ; .edr output, every 10 ps
nstlog         = 5000         ; .log output, every 10 ps
;
```

time step, total number of steps

frequency of outputs

```
cutoff-scheme  = Verlet
nstlist        = 20
rlist          = 1.2
vdwtype        = Cut-off
vdw-modifier    = Force-switch
rvdw_switch    = 1.0
rvdw           = 1.2
coulombtype    = PME
rcoulomb       = 1.2
;
```

some force field specific cutoffs for non-bonded interactions

```
tcoupl         = v-rescale
tc_grps        = Protlig Membrane Solvent
tau_t          = 1.0 1.0 1.0
ref_t          = 303.15 303.15 303.15
;
```

temperature and thermostat

```
pcoupl         = C-rescale
pcoupltype     = semiisotropic
tau_p          = 5.0
compressibility = 4.5e-5 4.5e-5
ref_p          = 1.0 1.0
;
```

pressure and barostat

```
constraints    = h-bonds
constraint_algorithm = LINCS
continuation    = yes
;
```

constraining C-H bonds

```
nstcomm        = 100
comm_mode      = linear
comm_grps      = Protlig_Membrane Solvent
;
```

removal of translational velocities

Running simulations: hardware

- High-performance computing (**HPC**) clusters
- MD simulations are efficiently run on **GPUs**



LUMI, Finland



Grendel, Aarhus (@ Physics)



Mare Nostrum, Barcelona

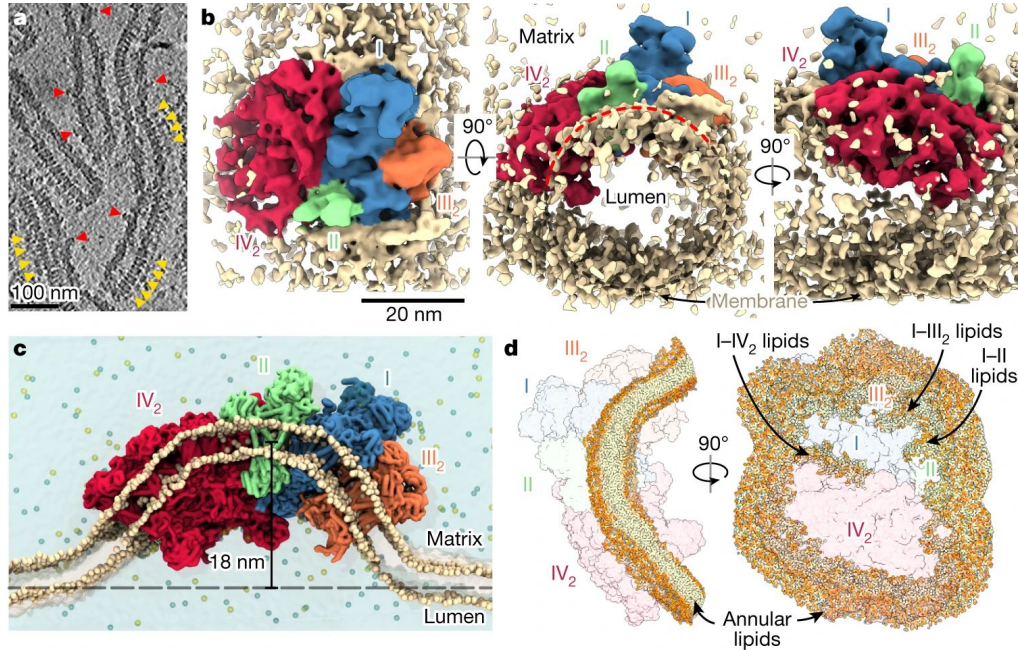


Folding@Home

We have simulations... Now what?

- Each simulation trajectory: 10++ Gb of data
- Information about movement of every atom in the system
 - Too much data
- Challenge: rationalising MD output
 - The approach will heavily depend on the research question that is being asked

Chapter 4: Applications

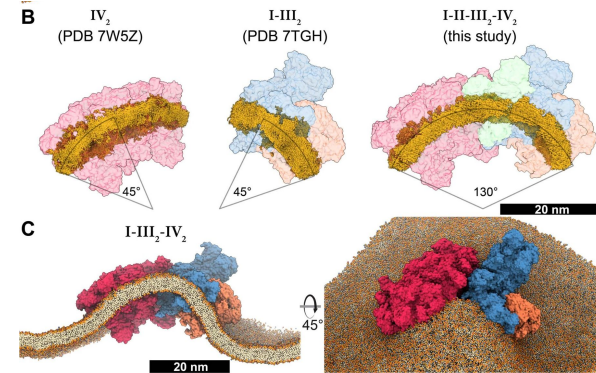


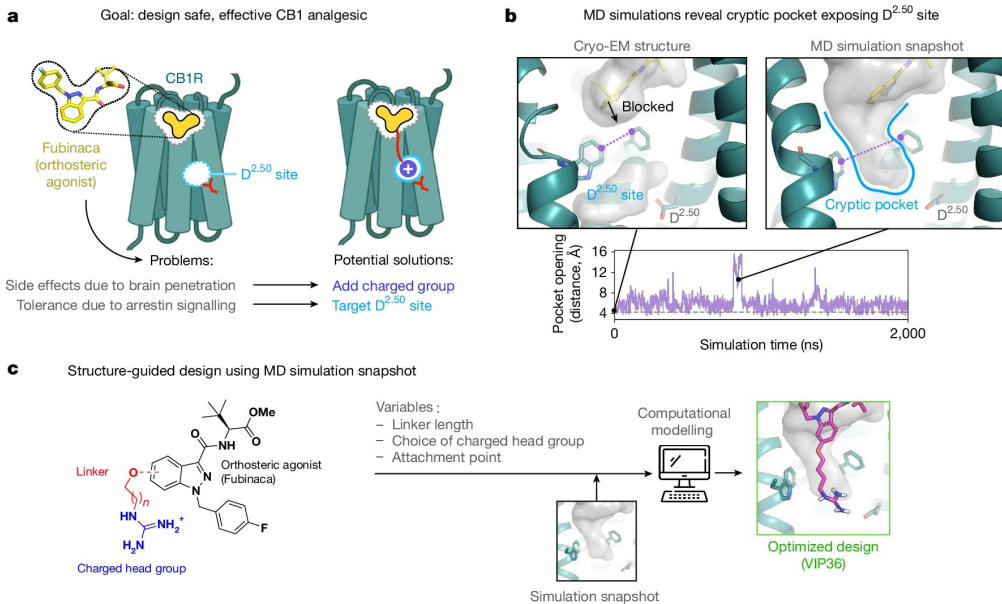
Mühleip, A. *et al. Nature* (2023).

<https://doi.org/10.1038/s41586-023-05817-y>

Structural basis of mitochondrial membrane bending by the I-II-III₂-IV₂ supercomplex

- Supercomplex induces membrane curvature





A cryptic pocket in CB1 drives peripheral and functional selectivity

- Cannabinoid receptor 1 (CB1) - non-opioid target for pain relief, but needs to be targeted **outside** the brain
- Detection of cryptic pockets using MD simulations
- Design of pocket-specific drugs that do not cross the brain-blood barrier

Rangari, V.A., *et al.* *Nature* (2025).

<https://doi.org/10.1038/s41586-025-08618-7>

Chapter 5: Past, present, future

How it started

Nature Vol. 267 16 June 1977

585

articles

Dynamics of folded proteins

J. Andrew McCammon, Bruce R. Gelin & Martin Karplus

Department of Chemistry, Harvard University, Cambridge, Massachusetts 02138

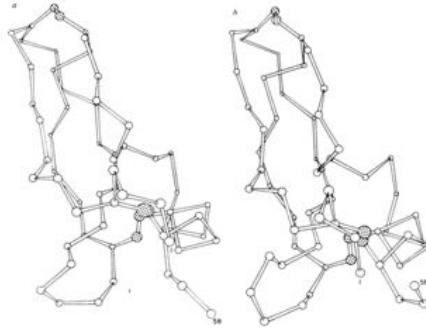
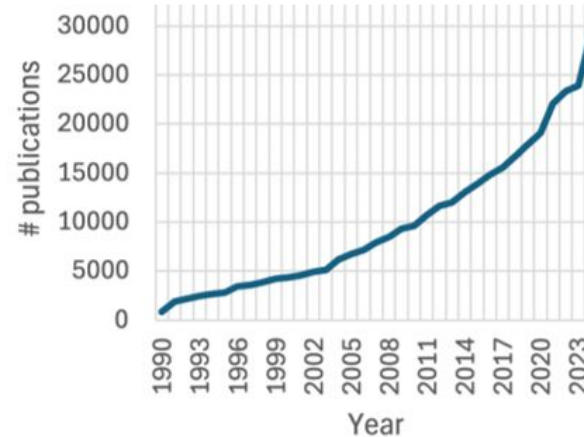


Fig. 1 The peptide backbone (in carbon) and disulfide bonds of P11. a, X-ray structure¹⁰. b, Time evolved structure after 3.2 ps of dynamical simulation.

How it is going



Publications per year with 'molecular dynamics' in title, abstract or keywords

The reasons behind it...

- The number of available structures has exploded
(+ predictions of protein structure via AlphaFold and co.)
- Computing power is increasing
- Simulation software is getting faster
- The method is continually developing

Nobel Prize in Chemistry 2013



© Nobel Prize Outreach.
Photo: A. Mahmoud

Martin Karplus

Prize share: 1/3



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Michael Levitt

Prize share: 1/3



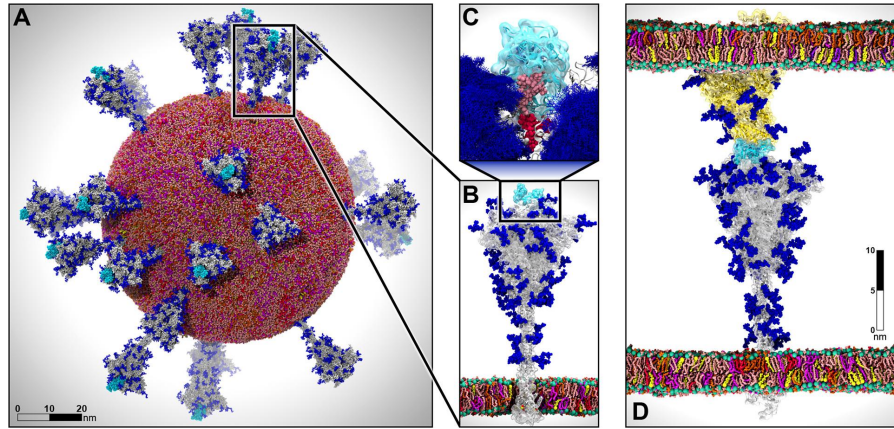
© Nobel Prize Outreach.
Photo: A. Mahmoud

Arieh Warshel

Prize share: 1/3

The Nobel Prize in Chemistry 2013 was awarded jointly to Martin Karplus, Michael Levitt and Arieh Warshel "for the development of multiscale models for complex chemical systems"

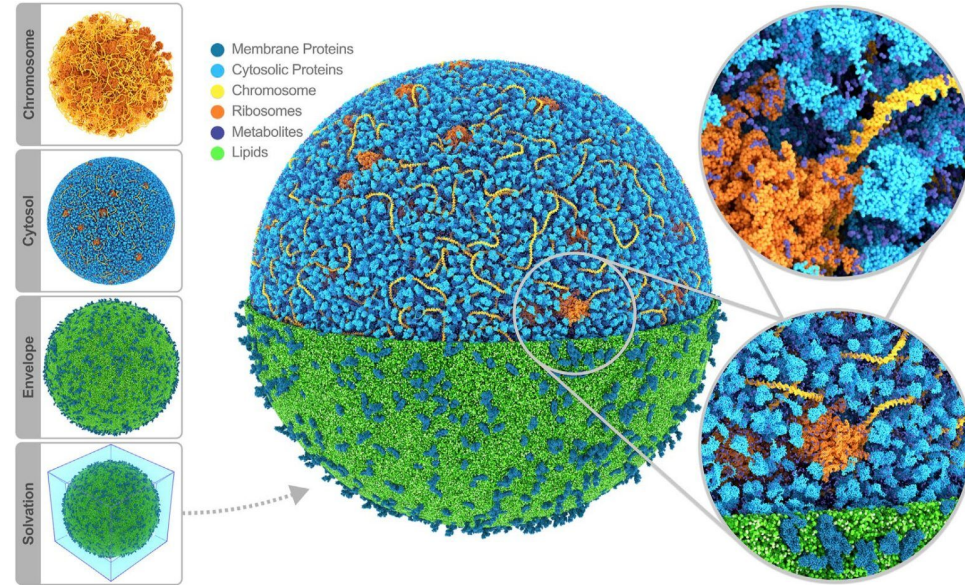
The systems are getting bigger...



Casalino et al. 2021

<https://doi.org/10.1177/1094342021100645>

Whole viral particle (SARS-CoV-2)



Stevens et al. 2023 <https://doi.org/10.3389/fchem.2023.1106495>

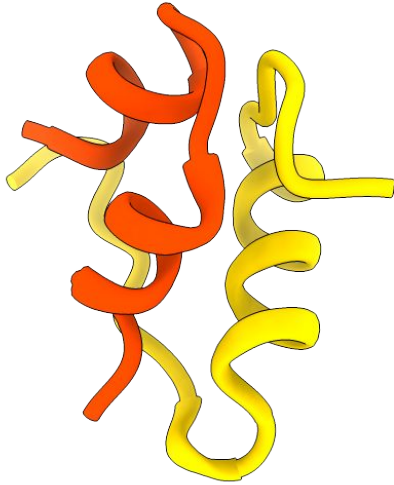
Minimal cell (JCVI-syn3A)

... but adequate sampling is still an issue!

Ergodic hypothesis states that a thermodynamic system will explore all states **if given enough time**.

- Bigger and more complex systems → more degrees of freedom → more sampling required
- Difficult to know how much sampling is “enough” sampling (cost-benefit analysis)
 - ✓ Sampling states around the energy minimum
 - × Sampling big conformational transitions → enhanced sampling methods
 - × Sampling very complex systems → coarse-grained methods

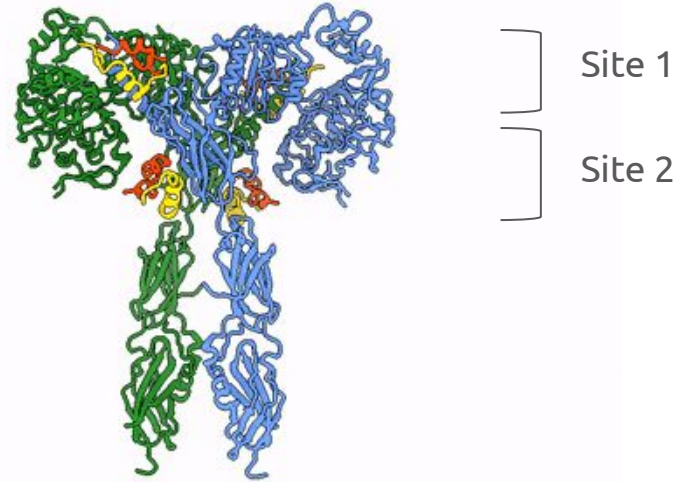
Chapter 6: Exercise



Insulin

“protein of interest”

- Insulin is a hormone regulating metabolism
- Promoting glucose uptake into the cell
- **Mechanism of action:** Insulin binds to the insulin receptor and activates it, which in turn starts the signalling cascade



Insulin receptor with bound insulins

Aims for tomorrow:

- compare the two conformations of insulin (originating from Site 1 and Site 2)
- compare the two MD simulations of insulin, where the starting conformations are different

Part 1: Visualisation



Part 2: Data analysis



What you will learn in the process:

- Visualising MD trajectories in PyMOL
- Performing some simple data analyses in Python
- Learning how to deal with too much data